

A Multi-Agent Ontology-based Architecture for Semantic Interoperability and Reasoning in Microgrid Energy Management

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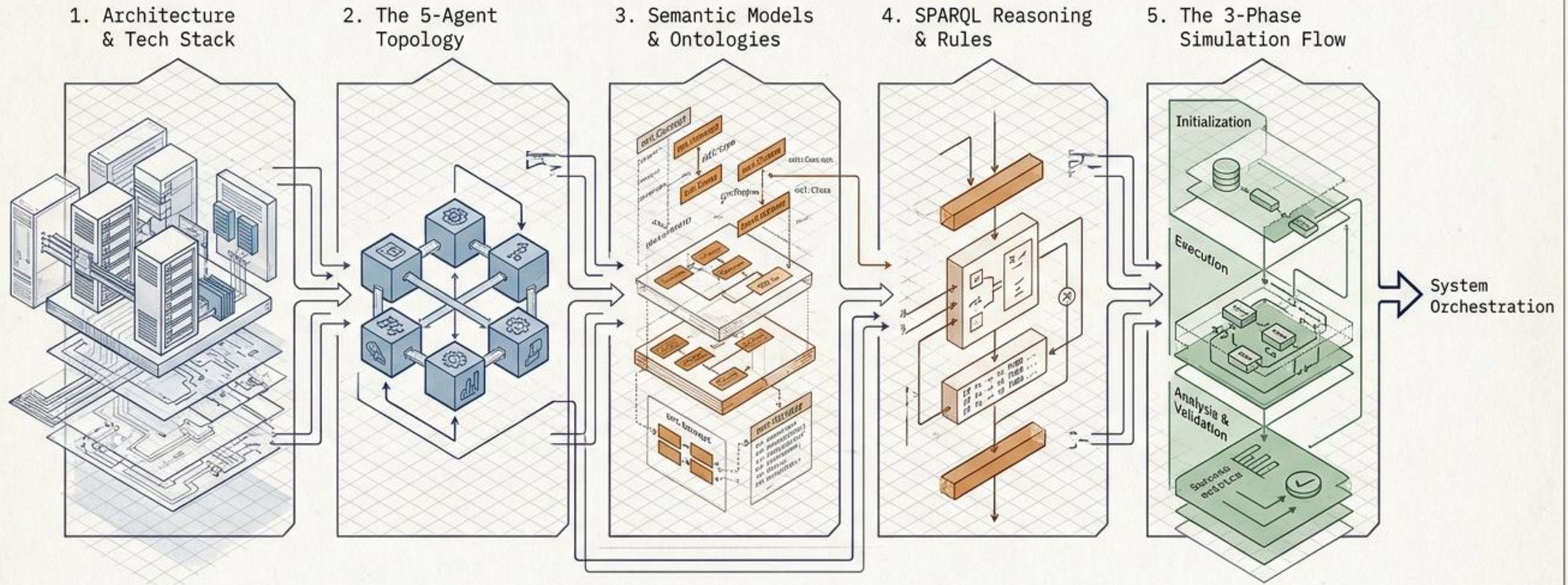
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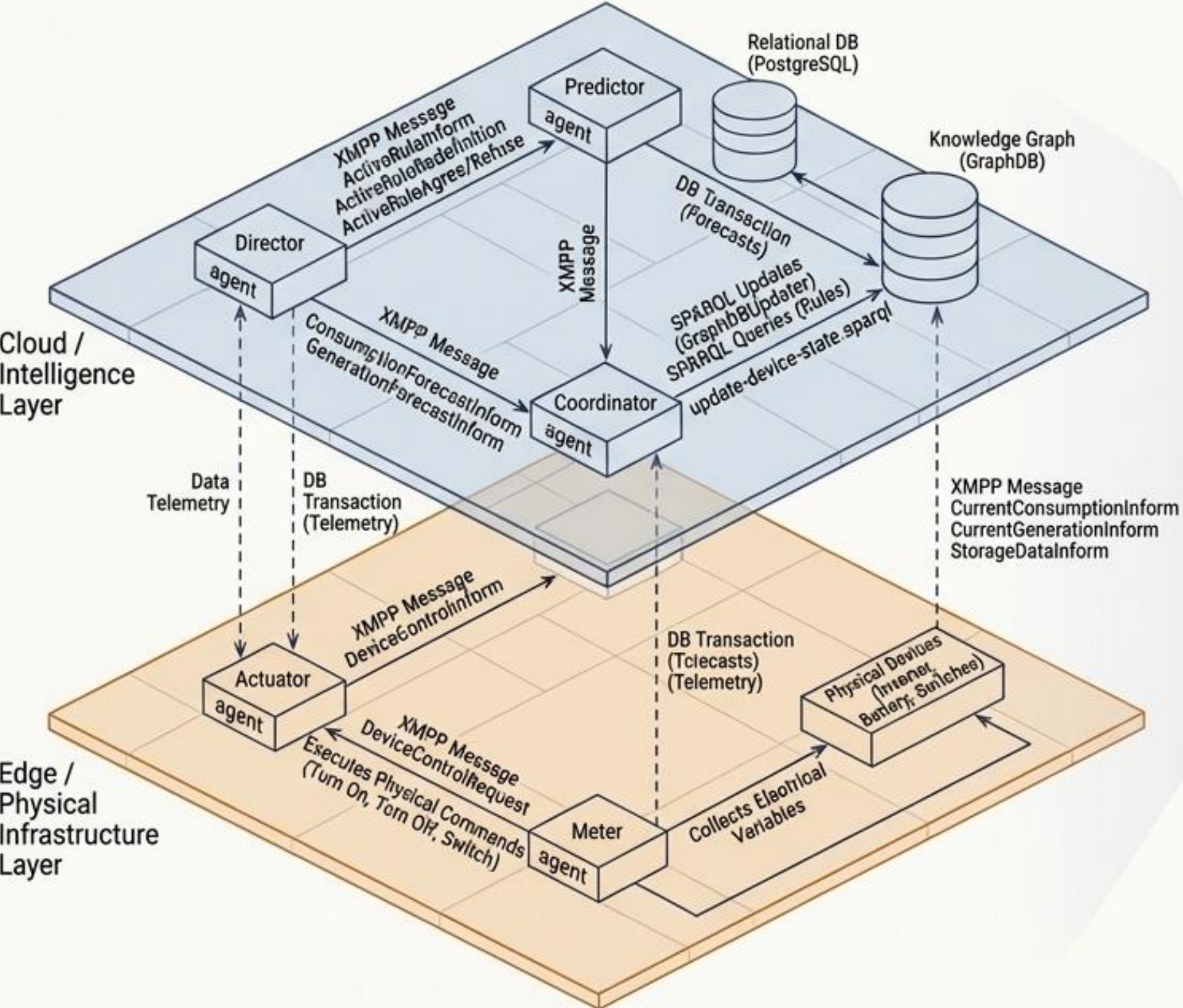
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Deconstructing the masIFRS Methodology



Segregating Intelligence from Infrastructure



1

Orchestration:

SPADE: Enables asynchronous agent actions and overall orchestration.

2

Messaging:

XMPP: Ensures decentralized, structured communication of control data and actuator signals.

3

Persistence:

PostgreSQL: Stores historical telemetry, time-series datasets, and ML projections.

4

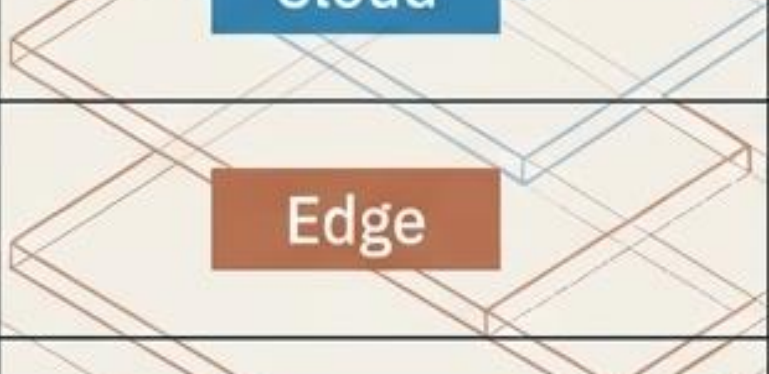
Semantics:

GraphDB: Acts as the semantic database managing domain ontology, validating runtime rules, and enabling logical inference via SPARQL.

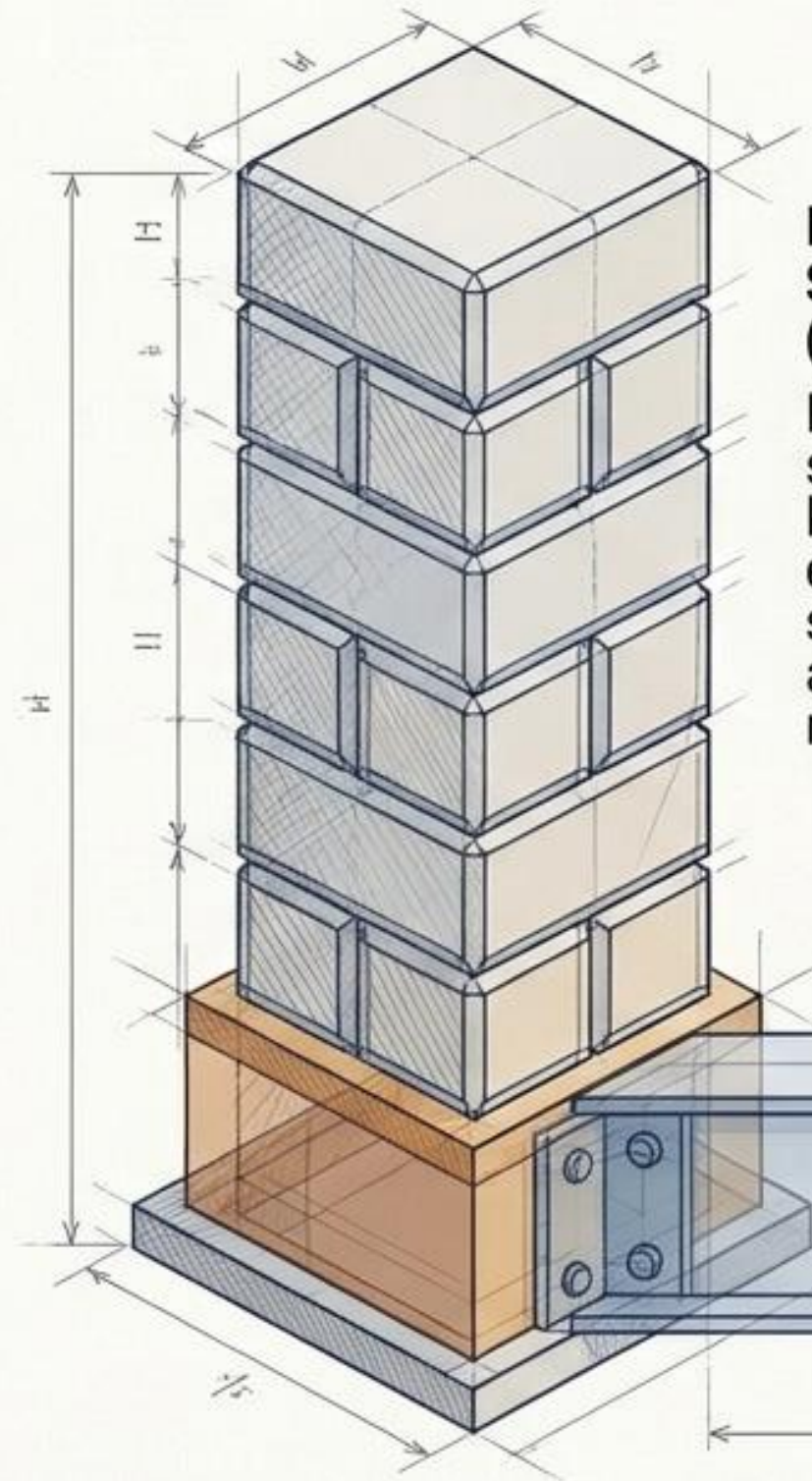
Tech Stack

The 5-Agent Operational Topology

Defining the decentralized autonomous agents managing the masIFRS microgrid.

	Environment	Core Function	Interaction Protocol
Director		Sets global policy & dynamically manages active rules	Accepts or rejects the application of the rules, and may suggest changes to them.
Coordinator		Orchestrates network, merges measurements, and executes SPARQL rules	Assesses consumption, generation, and prediction data to command acting agents
Predictor		Uses the ChronosForecaster ML model for 1-hour load/generation forecasting	Responds to requests with advance time-series estimates
Meter		Harvests raw telemetry from physical sensors	Formats raw measurements into standardized current load/capacity messages
Actuator		Translates logical decisions into physical commands	Executes on/off/switching commands on inverters and smart switches

Bridging Domain Knowledge with Interaction Protocols

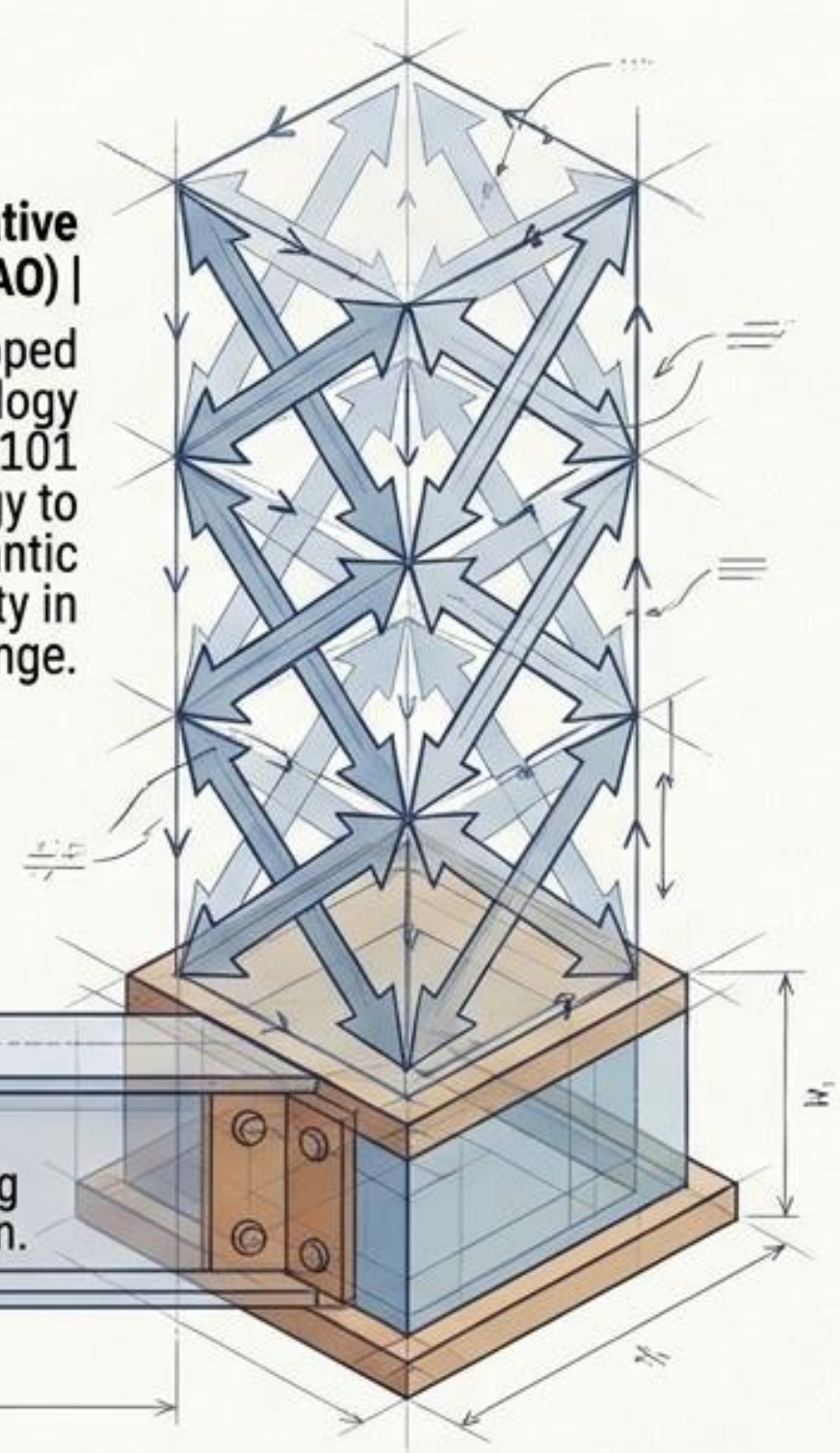


Intelligent Energy Systems Ontology (IESO) |

Reused to provide standardized domain knowledge for electrical power systems, smart grids, and physical device representation.

Communicative Acts Ontology (CAO) |

Custom-developed application ontology using the 101 methodology to ensure semantic interoperability in message exchange.



Architectural Goal: Avoid hardcoding logic. Operating rules live in the ontologies, enabling runtime updates without system recompilation.

Deconstructing the IESO Domain Ontology

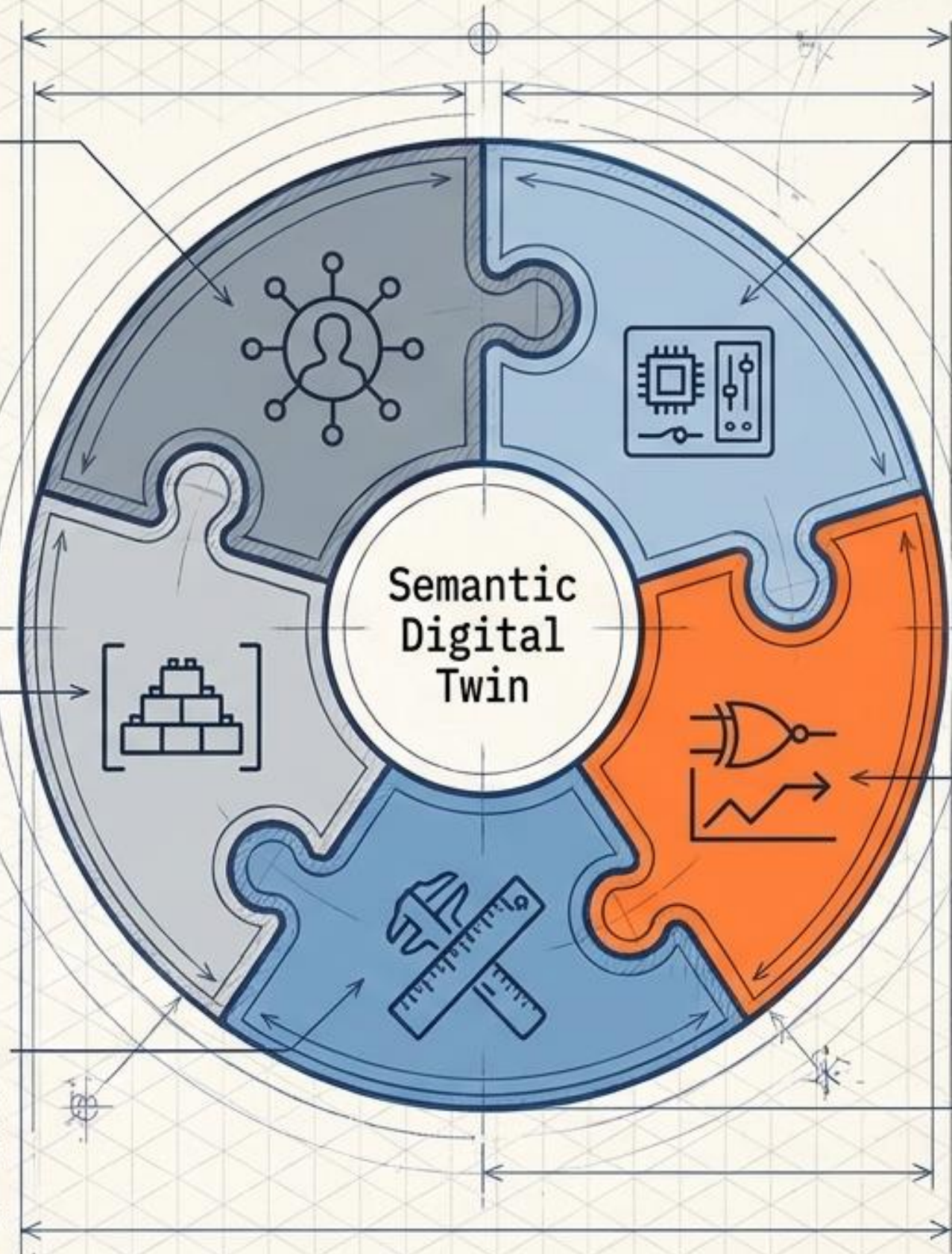
Filtering 14 comprehensive modules down to the 5 core pillars required to build an agile, functional Semantic Digital Twin.

Architectural Goal: Do not overcomplicate the model. By strictly utilizing these 5 modules, masIFRS achieves a perfectly aligned connective routing between digital logic and physical infrastructure.

Actor Roles
Instantiates stakeholders, defines agent identities, and outlines predetermined behavioral interactions.

Commons
Provides the foundational semantic concepts and overarching bridging structures for the domain.

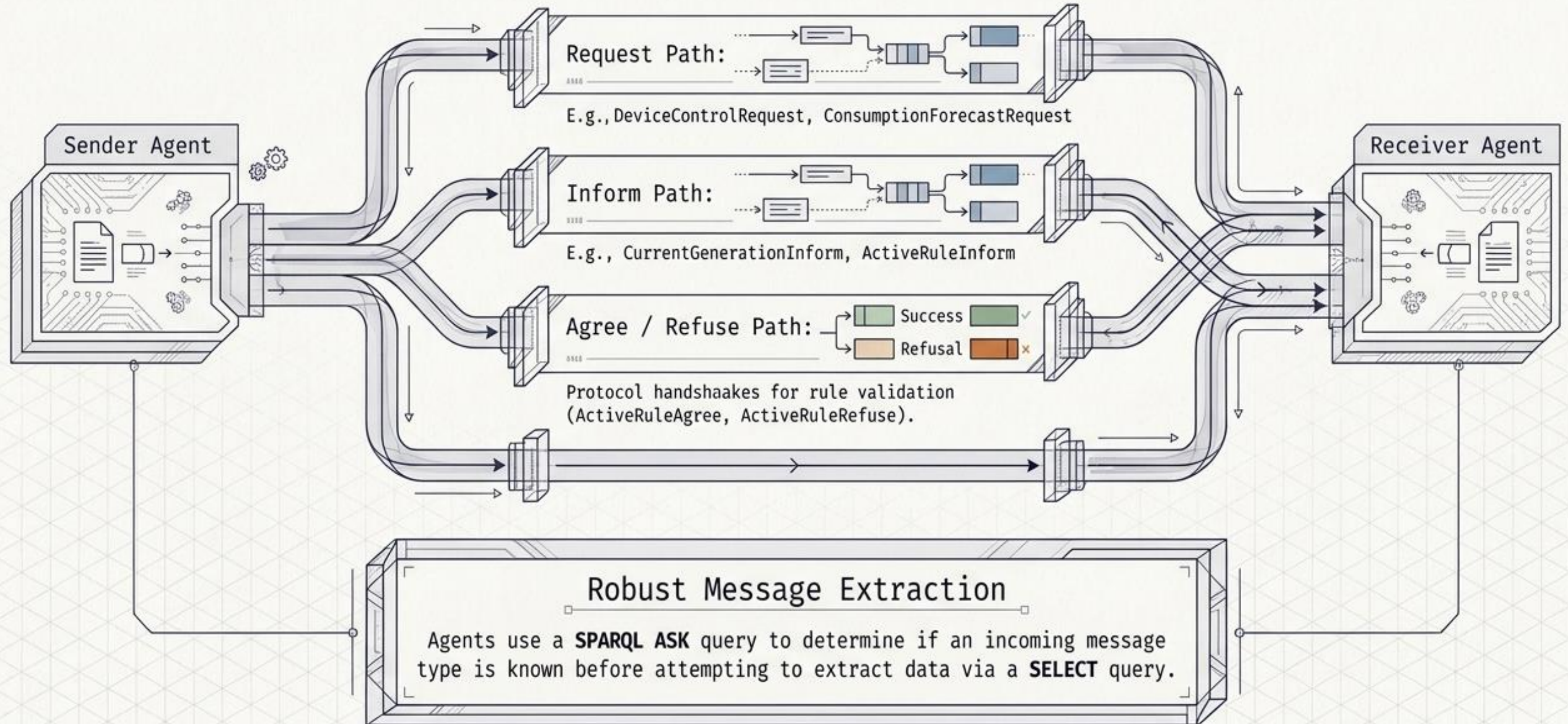
Measure
Standardizes SI units to seamlessly bridge forecasted models with edge sensor data, strictly preventing unit conversion errors.



Device
Semantically describes physical hardware, distinguishing between controllable and non-controllable assets, acceptable commands, and current states.

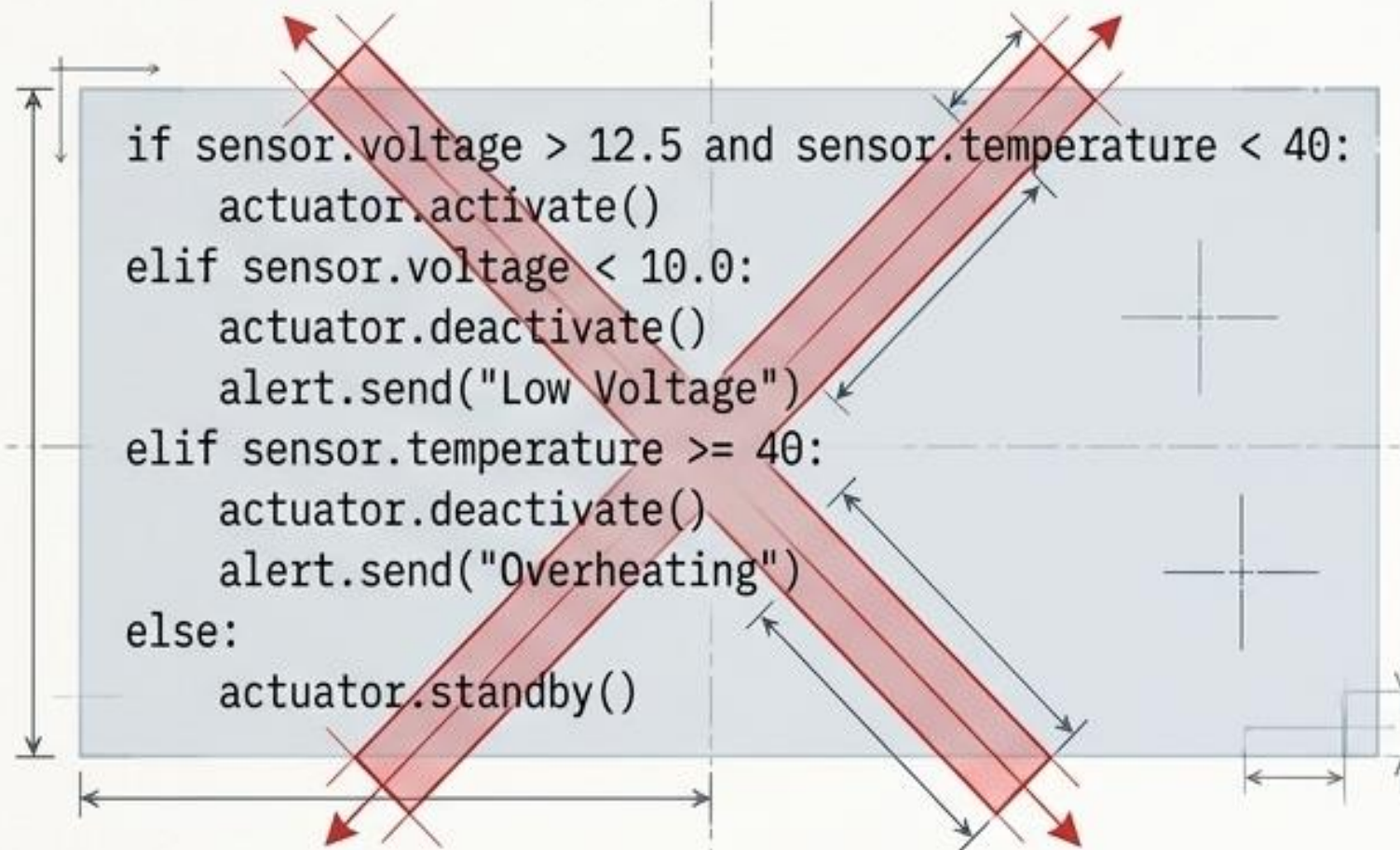
CRMP
Context-based Rules Matching Profile. The critical logic bridge. Defines operational conditions and actions as dynamic, executable logic graphs.

FIPA-Inspired Communicative Acts (CA0)



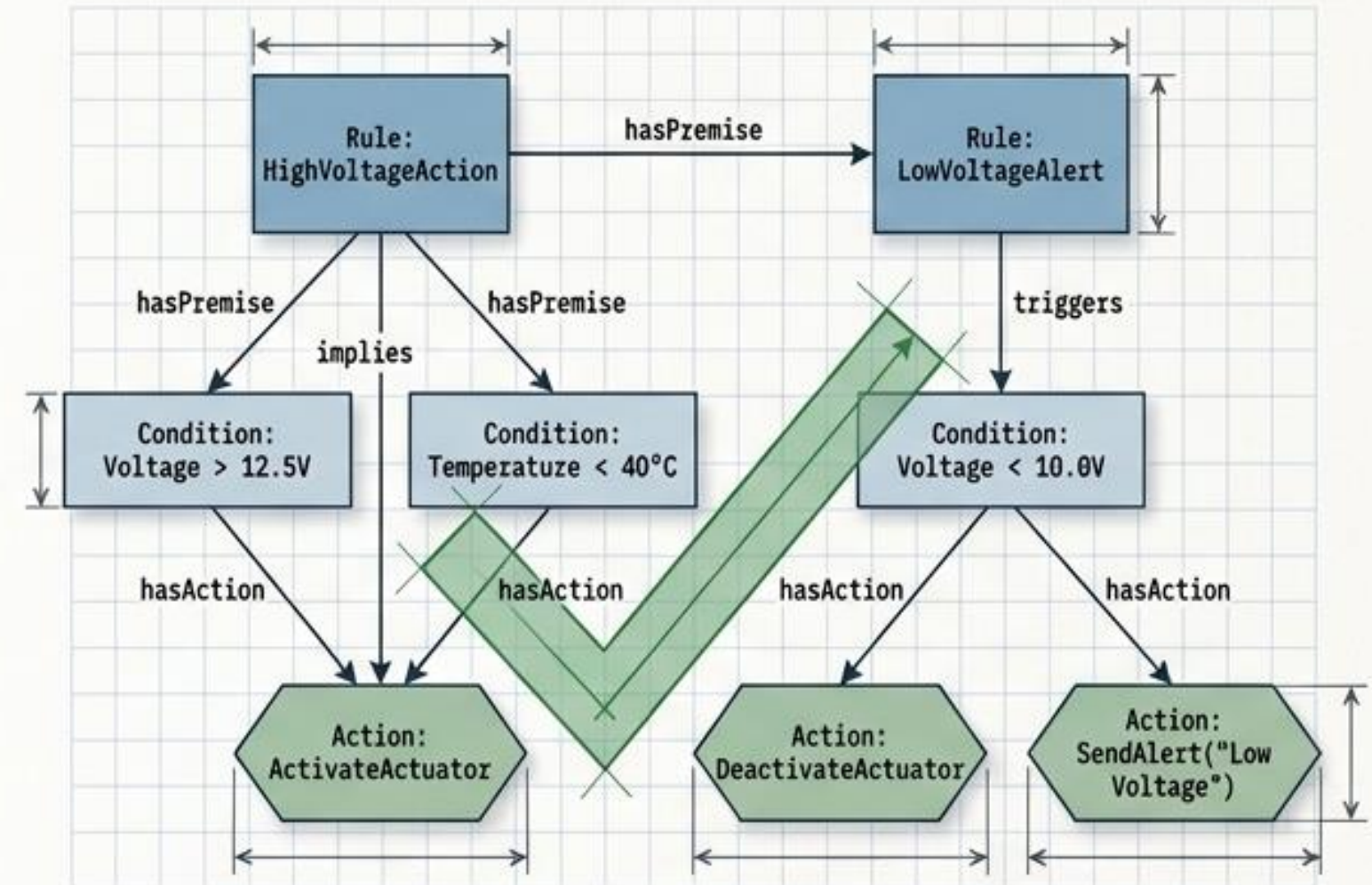
Semantic Elevation: Decoupling Logic from Implementation

Obsolete Method



Hardcoded Edge Logic (Rigid, requires recompilation, fragments decision-making).

masIFRS Method

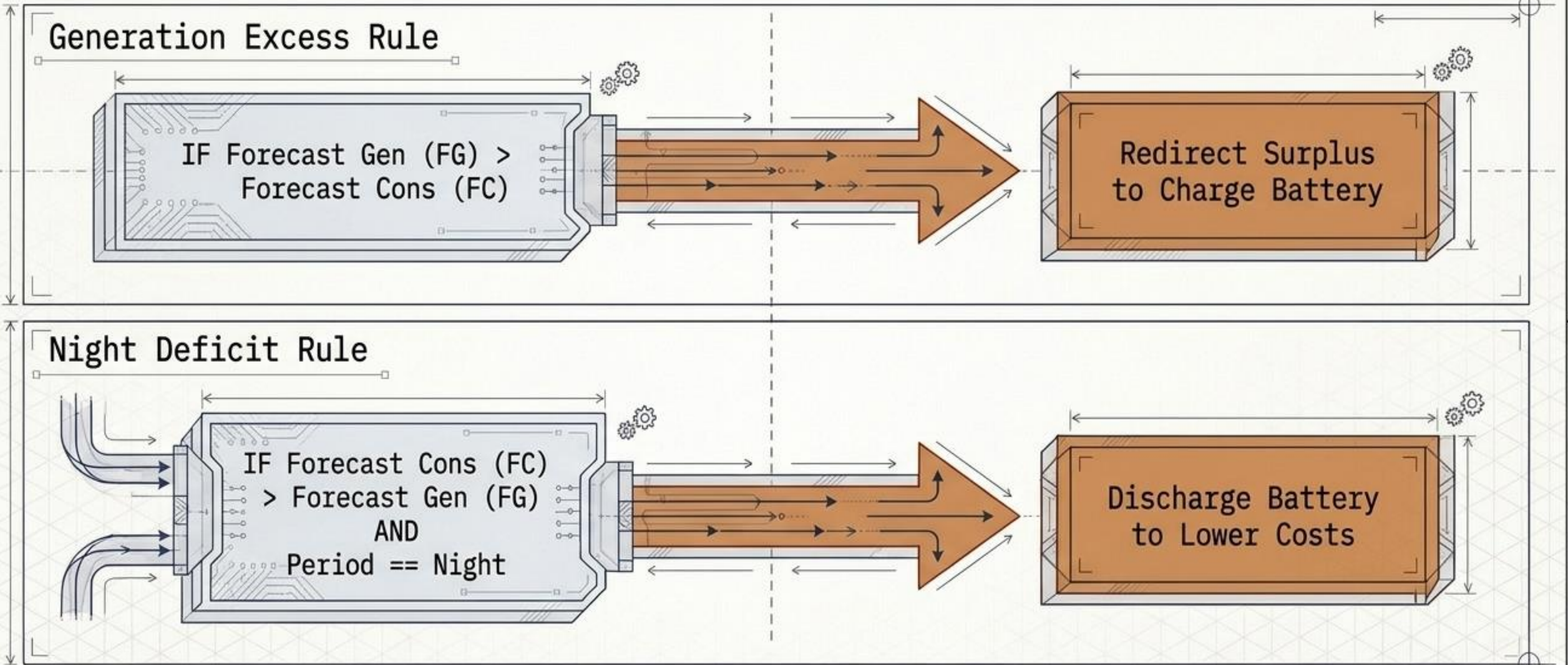


Rules-as-Graphs (Agile, runtime reconfigurable, centralized intelligence).

Core Concept:

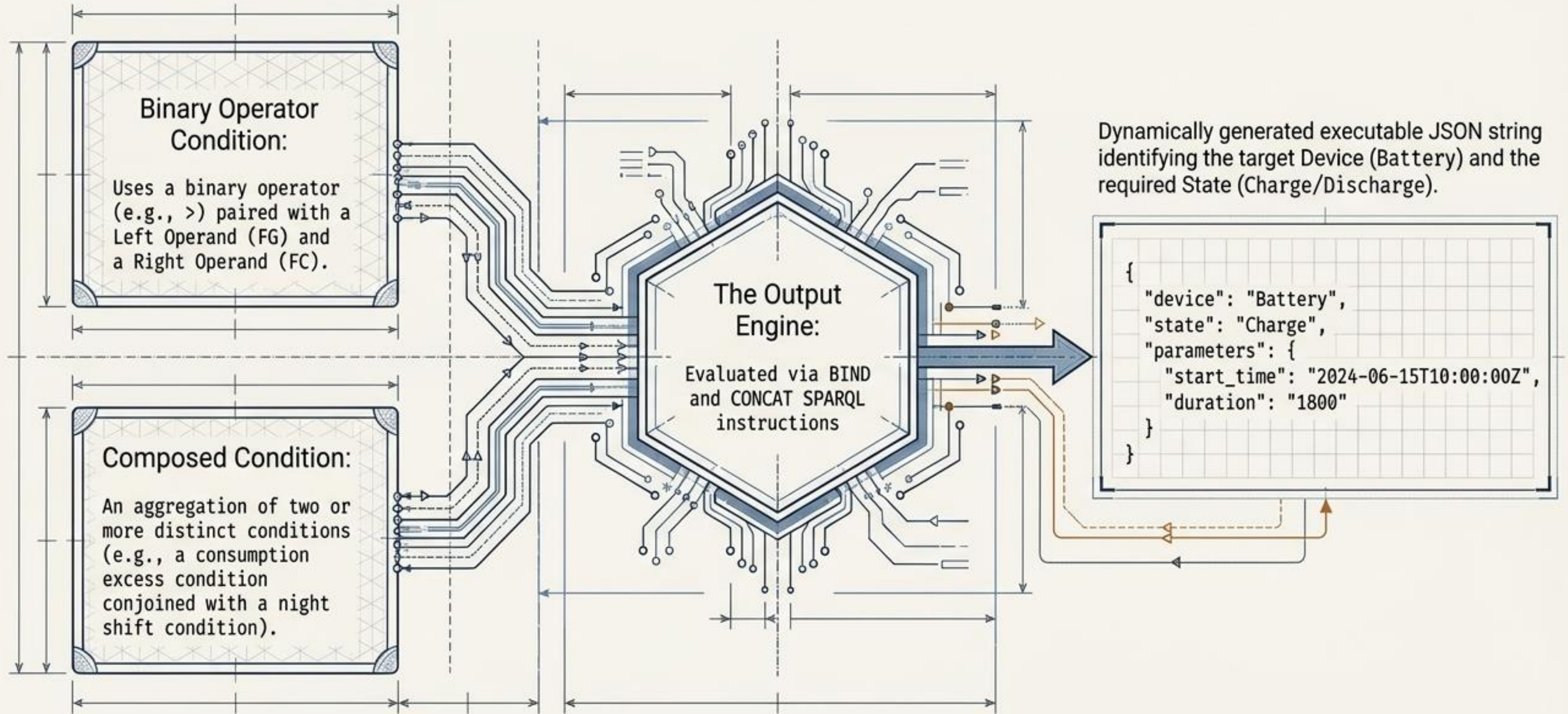
Policies are transformed into ontology individuals via the 'CRMP' module. These operational rules are evaluated dynamically using executable 'SPARQL' query templates.

Formulating the Operational Rules



These rules define whether energy will be stored, drawn from the battery, or drawn from the grid in the next execution period.

The Structural Anatomy of SPARQL Reasoning



Putting Architecture into Motion: The Orchestration Workflow

Simulation Environment

Campus Digital Twin:
Farroupilha University

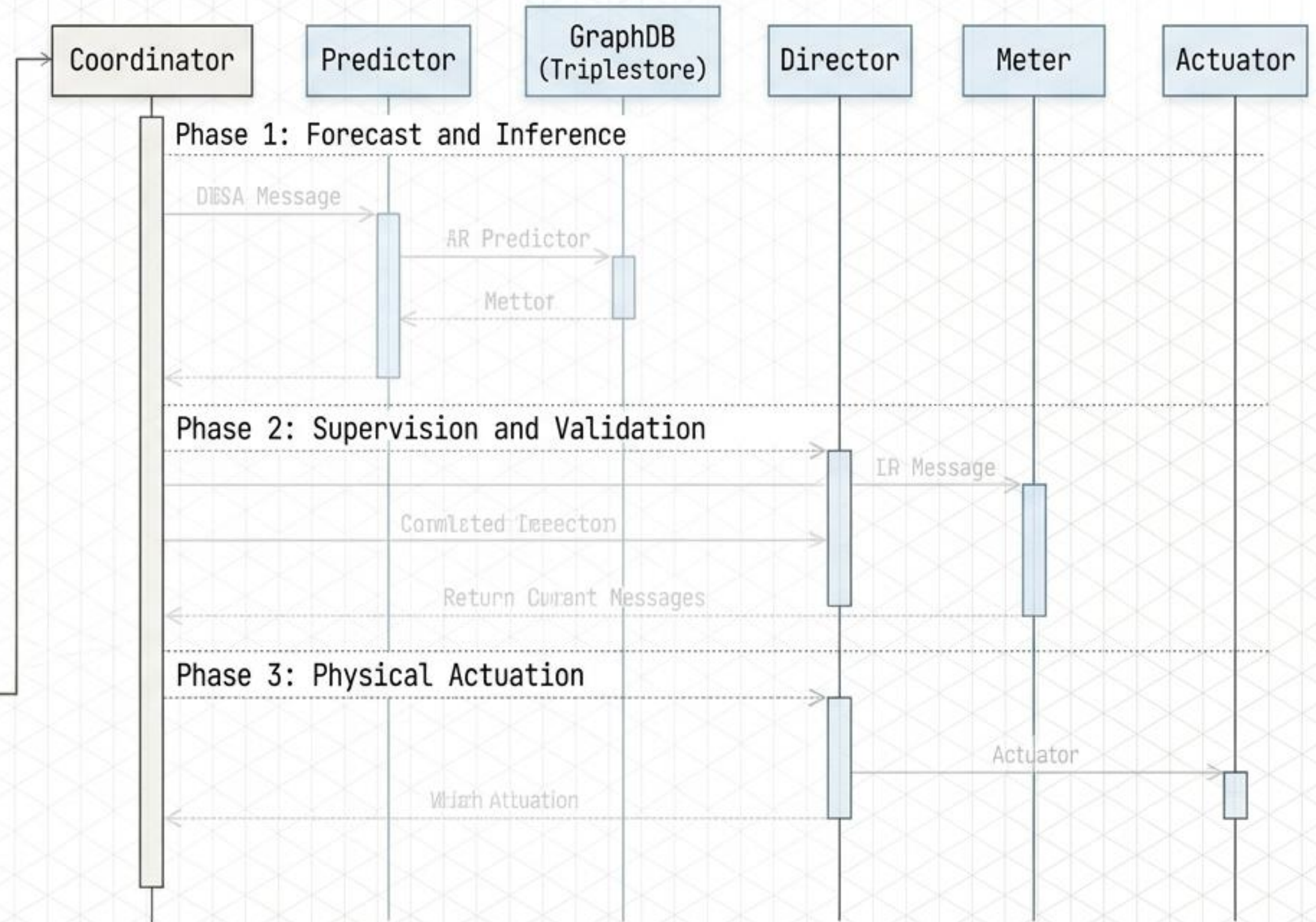
Photovoltaic Capacity:
60 kWp

Storage Capacity:
60 kWh

Load:
8 Classroom Buildings

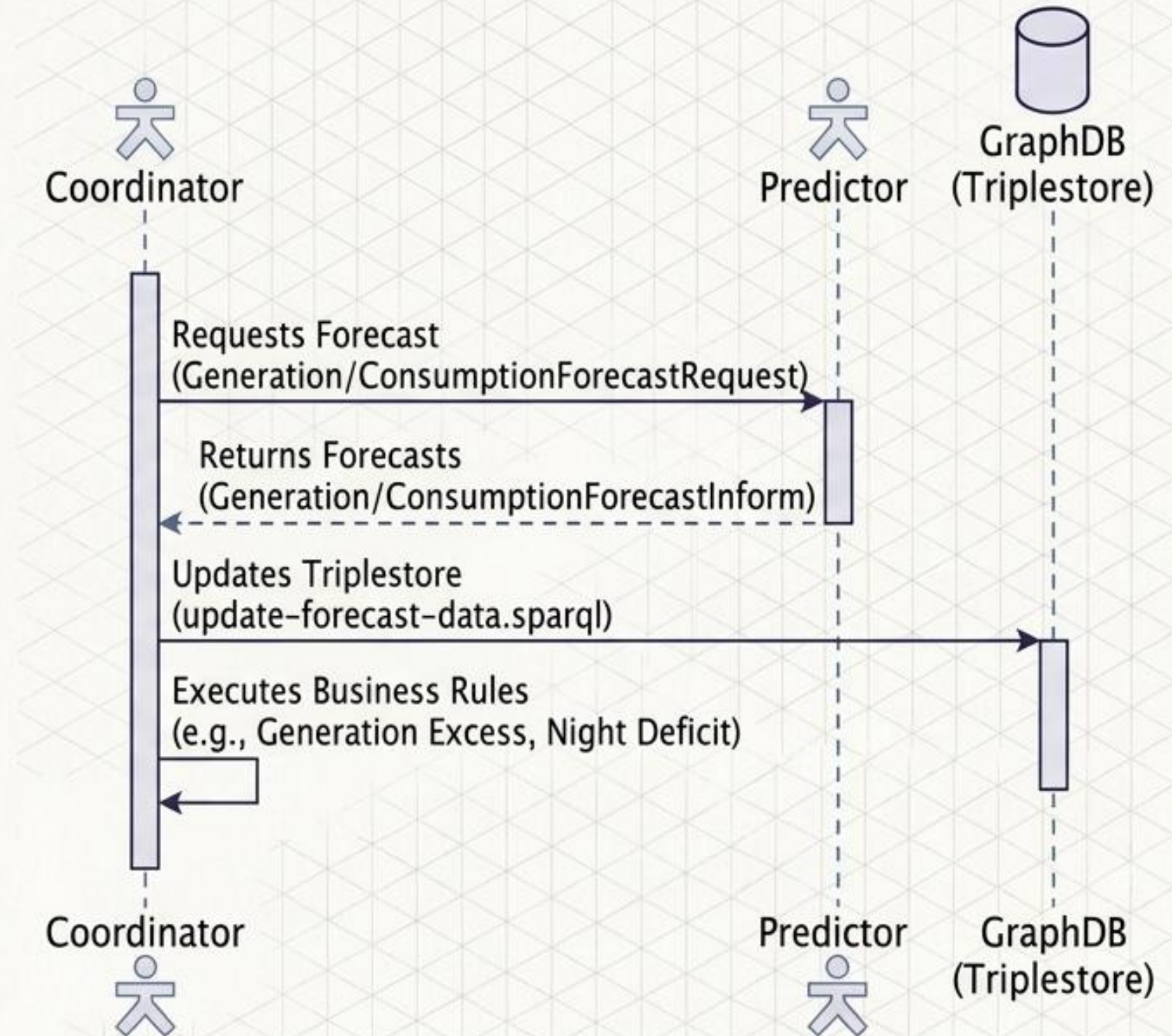
The Trigger:

The 3-Phase Simulation Cycle is triggered iteratively, specifically 5 minutes prior to the target execution period.



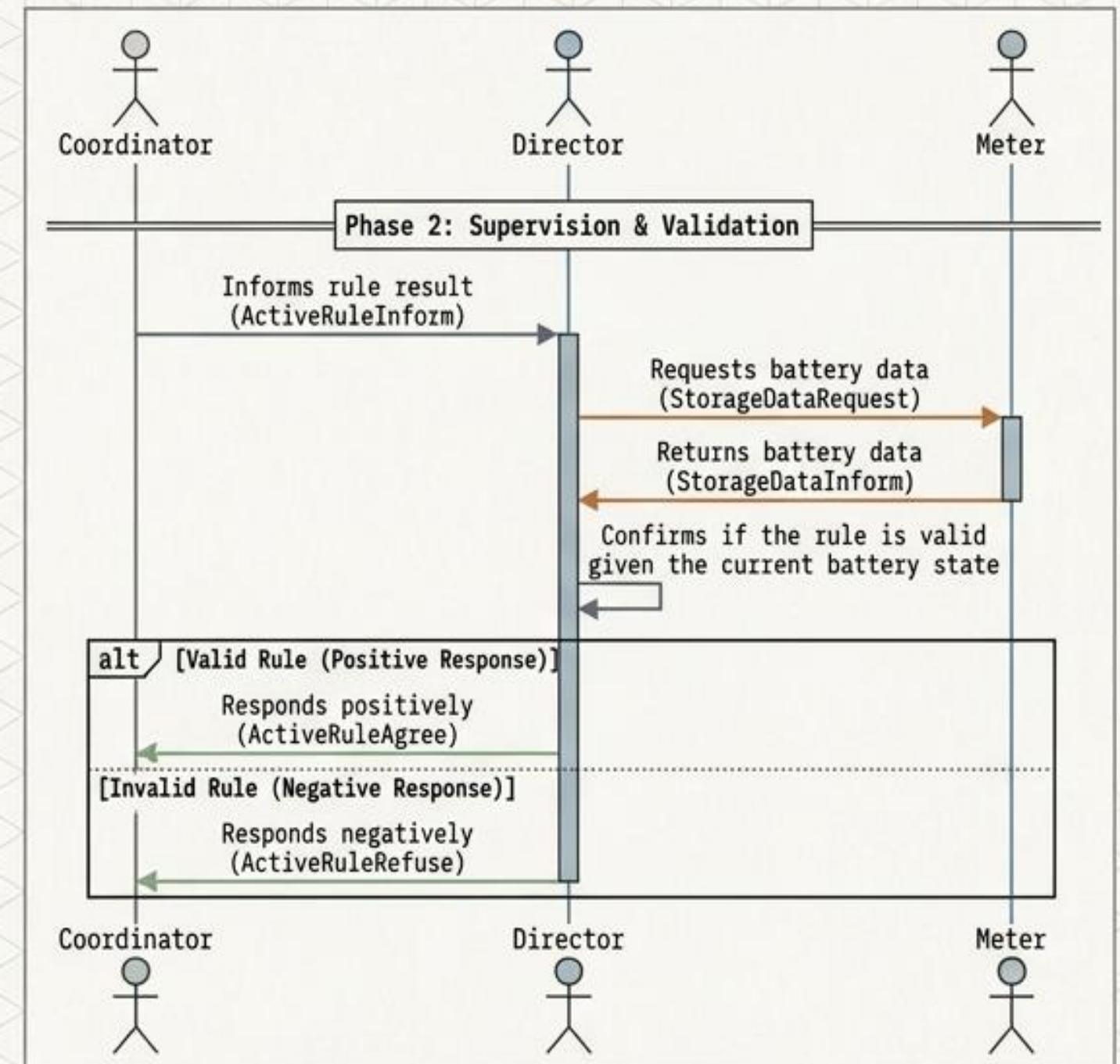
Phase 1: Forecast & Inference

1. Coordinator sends a ConsumptionForecastRequest and GenerationForecastRequest to the Predictor agent.
2. The Predictor ML engine responds with short-term projections.
3. The Coordinator leverages GraphDBUpdater to execute SPARQL Update statements, refreshing the triplestore with transient prediction data.
4. The Coordinator evaluates active CRMP business rules against the new data state.



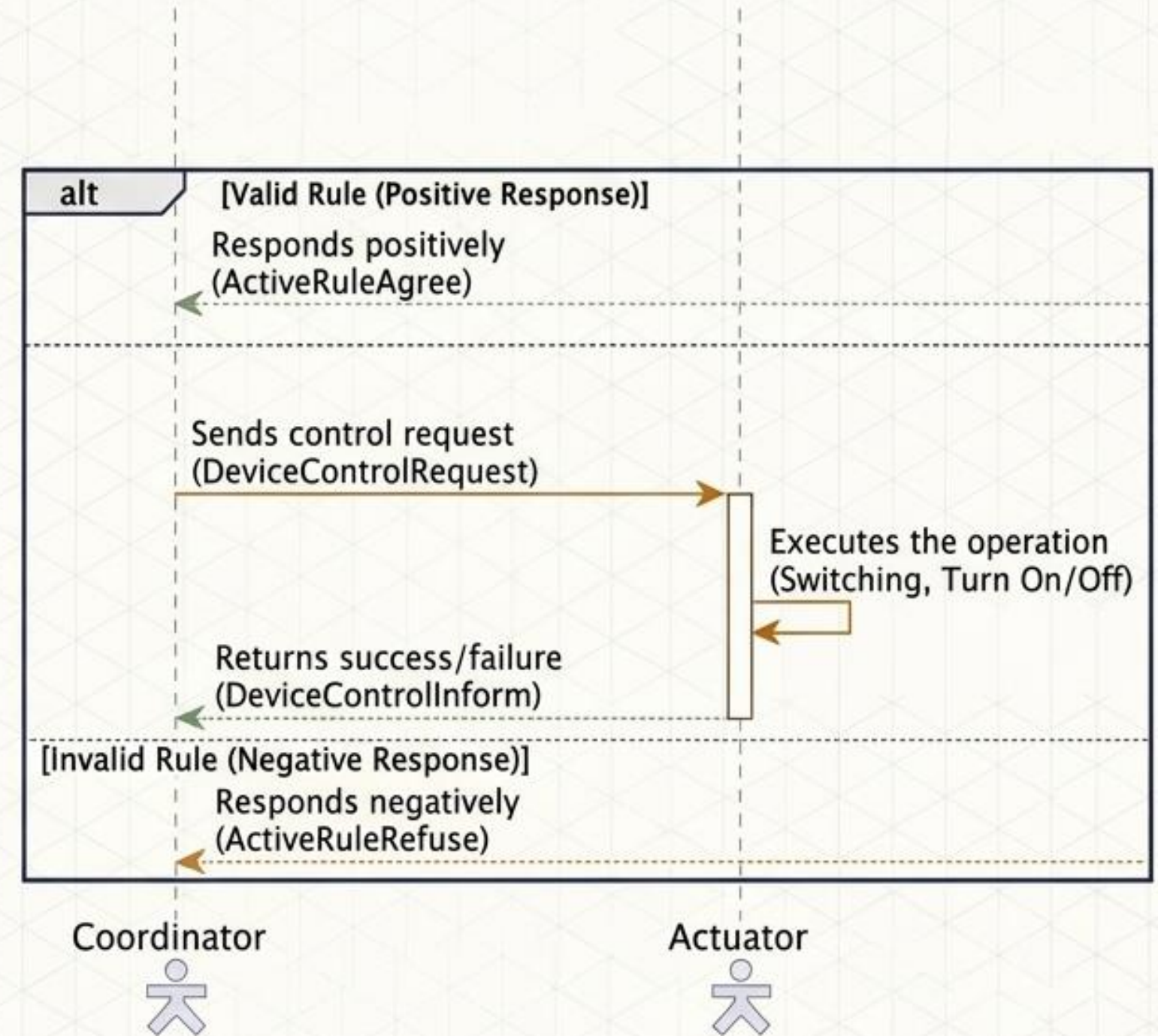
Phase 2: Supervision & Validation

1. The Coordinator sends an `ActiveRuleInform` to the Director.
2. The Director acts as the policy authority, querying the Meter via a `StorageDataRequest` to extract the immediate, real-time electrical state.
3. The Meter returns the physical battery limits (`StorageDataInform`).
4. The Director validates the theoretical rule against physical reality, returning either an `ActiveRuleAgree` or `ActiveRuleRefuse`.



Phase 3: Physical Actuation & Graph Update

1. Upon validation, the Coordinator sends the generated JSON command via a `DeviceControlRequest`.
2. The Actuator receives the request, parses the CAO intent, and triggers the physical hardware (e.g., switches, inverters).
3. The Actuator responds with a `DeviceControlInform` (Success/Failure).
4. The Semantic Digital Twin is updated to reflect the new, post-actuation physical reality, resetting the baseline for the next cycle.



Experimental Results & Microgrid Performance (Table 1)

CAMPUS SIMULATION PARAMETERS



60 kWp
Maximum Photovoltaic
Generation Capacity



60 kWh
Battery Storage Capacity



**8 Active Classroom
Buildings**
(March 25, 2025 data profile)

ENERGY SAVINGS & COST REDUCTION

-28.4%

Reduction in Grid Dependence / External Power Reliance

Optimized Battery Dispatch: Successfully shifting the microgrid from a reactive to a proactive state via predictive logic.

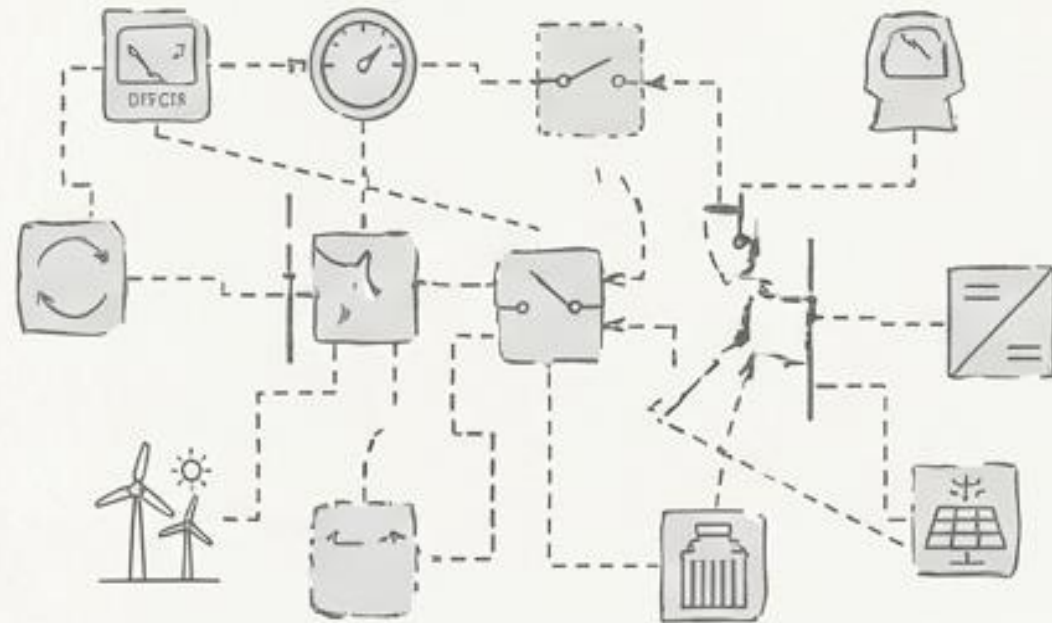
SEMANTIC ARCHITECTURE VIABILITY

**Real-time SPARQL reasoning response
times validated.**

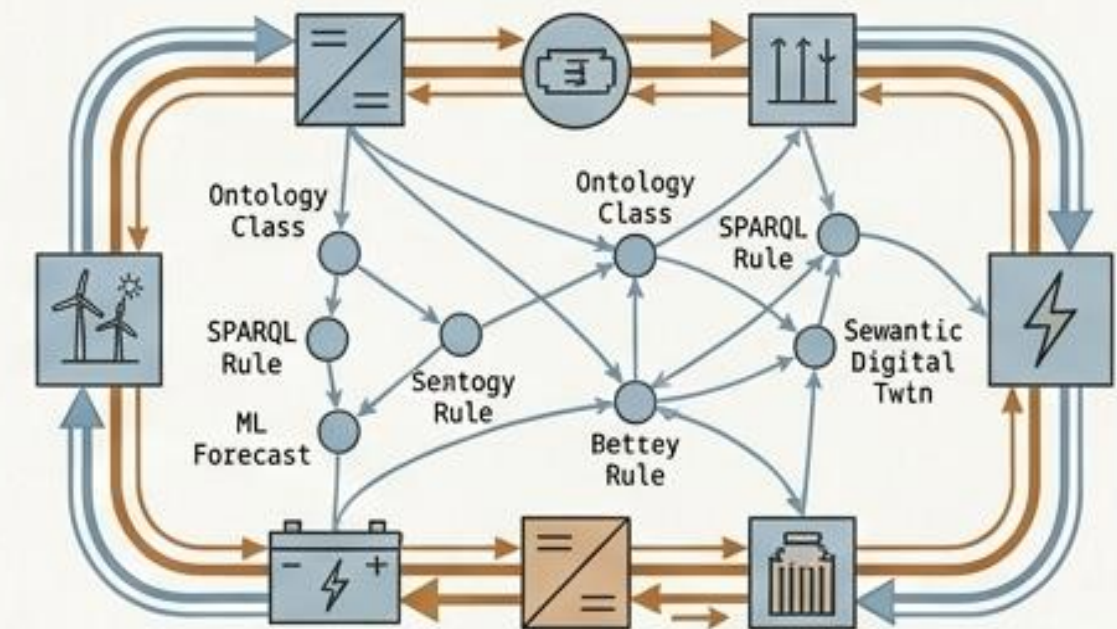
Proves the ontology-based MAS architecture is computationally viable for real-time, continuous Energy Management System (EMS) orchestration without latency bottlenecks.

The Impact: Realizing the Proactive Microgrid

Reactive Grid



Proactive Grid



Agnostic Control	Interoperability at the Edge	Proactive ML Integration
Software structure is decoupled from dispatch policies; operational logic is completely encapsulated in SPARQL graph rules.	IESO domain knowledge and custom CAO interactions provide seamless fault-tolerance across heterogeneous hardware.	ML forecasts are seamlessly elevated into semantic rules, replacing reactive hardware triggering with true predictive autonomy.

Questions & Discussion

Thank you for your attention. We welcome your questions, insights, and feedback.

Contact Matrix

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Build upon masIFRS

Access the complete Intelligent Energy Systems Ontology (IESO) modules, the custom Communicative Acts Ontology (CAO), the Python-based multi-agent framework, and our simulation datasets.

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```

```
files: communicative-acts.ttl, business-rules-kb.ttl  
business-smarties-acts.ttl
```



github.com/rmanfredini/masIFRS_AI

